Neural Layer Matrices and Activations

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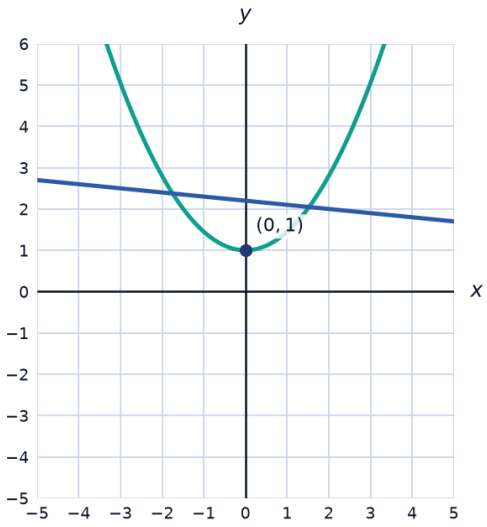
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Neural Layer Matrices & Activations Practice Exercises

[Matrices, Matrix Algebra & Inverses · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

2 [Foundational] • Target: Neural Layer Matrices & Activations Practice Exercises

Let A and B be invertible $n \times n$ matrices. Prove algebraically that: (a) $(AB)^T = B^T A^T$, and (b) $(AB)^{-1} = B^{-1}A^{-1}$.



3 [Foundational] • Target: Neural Layer Matrices & Activations Practice Exercises

Let $M_2(\mathbb{R})$ be the vector space of 2×2 real matrices. Determine, with formal verification via the Subspace Test, whether $W = \{A \in M_2(\mathbb{R}) : A^T = -A\}$ (the set of skew-symmetric matrices) is a subspace of $M_2(\mathbb{R})$. Find a basis and $\dim(W)$.

4 [Foundational] • Target: Neural Layer Matrices & Activations Practice Exercises

Express the invertible matrix $A = (1, 2; 3, 4)$ as a product of elementary matrices.

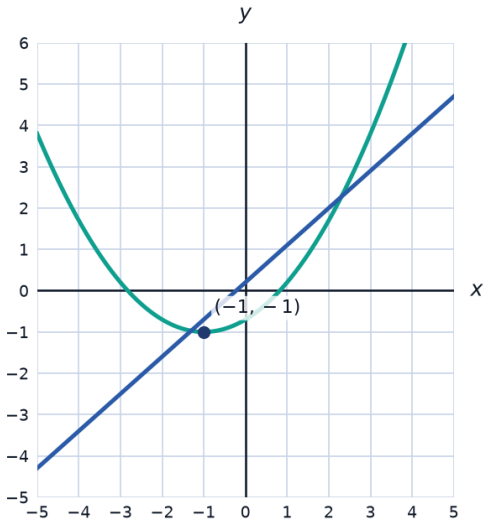
5 [Foundational] • Target: Neural Layer Matrices & Activations Practice Exercises

Practice Problem 2: Evaluate $\int_C \mathbf{F} \cdot d\mathbf{r}$ where $\mathbf{F} = abla(x^3y^2 + \sin(xz))$ and C is the helix $\mathbf{r}(t) = \langle \cos t, \sin t, t \rangle$ for $0 \le t \le 2\pi$.

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6 [Foundational] • Target: Neural Layer Matrices & Activations Practice Exercises

Level 3 · Advanced / University RigorSuppose an $n \times n$ matrix A satisfies the polynomial equation $A^3 - 4A^2 + 2A - 5I = 0$. Prove that A is invertible, and express A^{-1} as an explicit polynomial in A.



7 [Foundational] • Target: Neural Layer Matrices & Activations Practice Exercises

Practice Problem 2: Find the mass of a solid right circular cylinder of radius R and height h whose density at any point is proportional to its distance from the axis of symmetry, $\delta(r, \theta, z) = kr$ for constant $k > 0$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Neural Layer Matrices & Activations Practice Exercises

Practice Problem 1: Evaluate $\iiint_E (x^2 + y^2 + z^2)\, dV$, where E is the solid ball $x^2 + y^2 + z^2 \le a^2$.

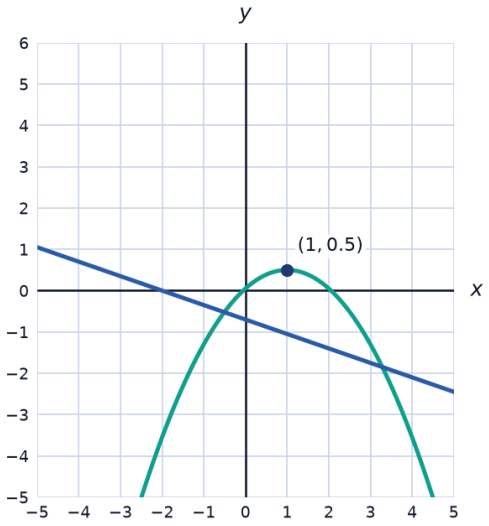
9 [Intermediate] • Target: Neural Layer Matrices & Activations Practice Exercises

Level 1 · FoundationsLet $M_{2\times 2}$ be the vector space of all 2×2 real matrices. Show that the set of all symmetric matrices $W = \{A \in M_{2\times 2} : A^T = A\}$ is a subspace of $M_{2\times 2}$.

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10 [Intermediate] • Target: Neural Layer Matrices & Activations Practice Exercises

Level 2 · MediumLet A and B be 3×3 matrices with $\det(A) = 3$ and $\det(B) = -2$. Compute the exact numerical value of: $\det(2A^2B^{-1}A^T)$.



11 [Intermediate] • Target: Neural Layer Matrices & Activations Practice Exercises

Level 2 · MediumLet $A = (3, 5; 1, 2)$ and $\mathbf{b} = (4; 1)$. Compute A^{-1} and use it to solve the linear system $A\mathbf{x} = \mathbf{b}$.

12 [Intermediate] • Target: Neural Layer Matrices & Activations Practice Exercises

Level 1 · FoundationsLet $A = (2, -1, 3; 4, 0, -2)$ and $\mathbf{x} = (3; -2; 1)$. Compute $A\mathbf{x}$ explicitly using the linear combination of columns definition.

13 [Intermediate] • Target: Neural Layer Matrices & Activations Practice Exercises

[Determinants & Geometric Volume · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

14 [Intermediate] • Target: Neural Layer Matrices & Activations Practice Exercises

[Eigenvalues, Eigenvectors & Diagonalization · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

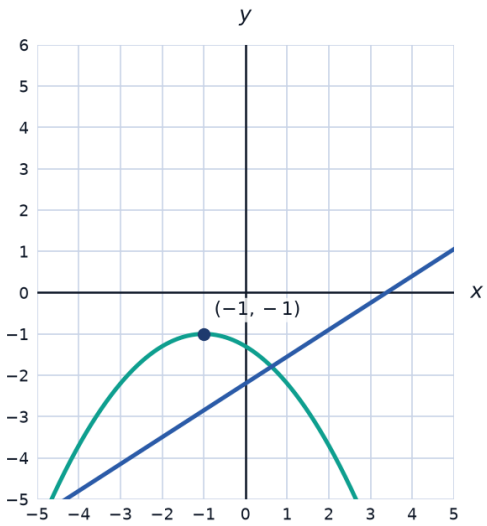
15 [Challenging] • Target: Neural Layer Matrices & Activations Practice Exercises • Cross-Lesson: 3.01 Gradient Descent and Optimization

[Systems of Linear Equations & Row Reduction · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

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16 **[Challenging] • Target: Neural Layer Matrices & Activations Practice Exercises • Cross-Lesson: 4.01 Conditional Probability and Bayes Rule**

[Vector Spaces, Subspaces & The Four Fundamental Subspaces · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.



17 **[Challenging] • Target: Neural Layer Matrices & Activations Practice Exercises • Cross-Lesson: 5.01 Distributions and Maximum Likelihood**

[Applications of Integration · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

18 **[Challenging] • Target: Neural Layer Matrices & Activations Practice Exercises • Cross-Lesson: 6.01 Linear Regression and OLS**

[Applications of Derivatives · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

19 **[Challenging] • Target: Neural Layer Matrices & Activations Practice Exercises • Cross-Lesson: 7.01 Logistic Regression and Classification**

[Convergence Tests & Power Series · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

20 **[Challenging] • Target: Neural Layer Matrices & Activations Practice Exercises • Cross-Lesson: 8.01 PCA and Dimensionality Reduction**

[Derivatives · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Neural Layer Matrices & Activations Practice Exercises

A fully-connected dense layer maps an input \(\mathbf{x} \in \mathbb{R}^{10} \) to \(\mathbf{y} \in \mathbb{R}^{5} \) using \(\mathbf{y} = \mathbf{W}\mathbf{x} + \mathbf{b} \). How many total learnable parameters are in this layer?

a. \(55 \)

b. \(50 \)

c. \(15 \)

d. \(65 \)

22 [Foundational] • Target: Neural Layer Matrices & Activations Practice Exercises

Evaluate the output of the standard ReLU activation function \(f(z) = \max(0, z) \) for inputs \(z_1 = -2 \) and \(z_2 = 3 \).

a. \(f(z_1) = 2, \; f(z_2) = 3 \)

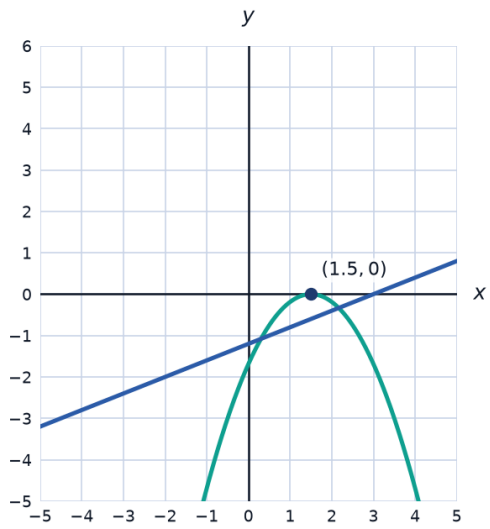
b. \(f(z_1) = -2, \; f(z_2) = 3 \)

c. \(f(z_1) = 0, \; f(z_2) = 0 \)

d. \(f(z_1) = 0, \; f(z_2) = 3 \)

23 [Foundational] • Target: Neural Layer Matrices & Activations Practice Exercises

Consider a linear layer with a weight matrix W and an input vector $\mathbf{x}$. If $W = (1, 2; 3, 4)$ and $\mathbf{x} = (5; 6)$, compute the output vector $\mathbf{y} = W\mathbf{x}$.



a. $(17; 39)$

b. $(11; 19)$

c. $(17; 19)$

d. $(11; 39)$

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24 [Foundational] • Target: Neural Layer Matrices & Activations Practice Exercises

Consider a linear transformation represented by the matrix $M = [2, k; 1, 3]$. A vector $\mathbf{v} = [1; 2]$ is transformed by M into $\mathbf{v}' = M\mathbf{v} = [8; 7]$. Determine the value of the unknown parameter k.

a. 1

b. 2

c. 3

d. 4

25 [Foundational] • Target: Neural Layer Matrices & Activations Practice Exercises

A linear transformation is defined by the matrix $M = (2, 1; 3, 4)$. If an input vector is $v = (x; 1)$, and the output vector after the transformation is $w = (5; 7)$, what is the value of x?

a. 1

b. 2

c. 3

d. 4

26 [Foundational] • Target: Neural Layer Matrices & Activations Practice Exercises

Consider a 2D dataset represented by the points $A = (1, 2)$, $B = (4, 2)$, and $C = (2, 5)$. If this dataset undergoes a linear transformation represented by the matrix $M = (2, 0; 0, 1)$, what are the coordinates of the transformed points A', B', and C'?

a. $A' = (2, 2), B' = (8, 2), C' = (4, 5)$

b. $A' = (1, 4), B' = (4, 4), C' = (2, 10)$

c. $A' = (2, 4), B' = (8, 4), C' = (4, 10)$

d. $A' = (3, 2), B' = (6, 2), C' = (4, 7)$

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27 [Foundational] • Target: Neural Layer Matrices & Activations Practice Exercises

Consider a dataset with n data points $\{\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_n\}$, where each $\mathbf{x}_i \in \mathbb{R}^p$. The sample covariance matrix is given by $\mathbf{S} = \frac{1}{n-1}\sum_{i=1}^{n}(\mathbf{x}_i - \bar{\mathbf{x}})(\mathbf{x}_i - \bar{\mathbf{x}})^T$, where $\bar{\mathbf{x}} = \frac{1}{n}\sum_{i=1}^{n}\mathbf{x}_i$ is the sample mean. Suppose PCA is performed on this dataset, resulting in principal components defined by the eigenvectors of $\mathbf{S}$ and corresponding variances given by the eigenvalues. Let $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_p$ be the orthonormal eigenvectors of $\mathbf{S}$ with associated eigenvalues $\lambda_1 \geq \lambda_2 \geq \ldots \geq \lambda_p$. The first principal component captures the direction of maximum variance, λ_1, and the second principal component captures the next largest variance, λ_2, orthogonal to the first. If we decide to reduce the dimensionality from p to k by keeping only the first k principal components, the proportion of the total variance retained is given by $\frac{\sum_{i=1}^{k}\lambda_i}{\sum_{i=1}^{p}\lambda_i}$. Suppose for a particular dataset, the eigenvalues of the covariance matrix are $\{10.5, 5.2, 2.1, 0.8, 0.4\}$. If we aim to retain at least 90% of the total variance, what is the minimum number of principal components required?

a. 3

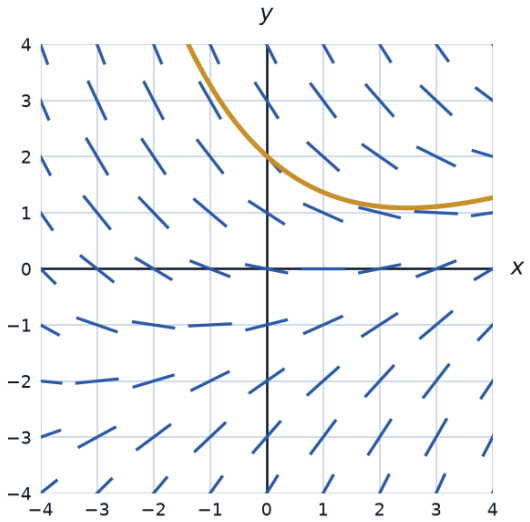
b. 2

c. 4

d. 5

28 [Foundational] • Target: Neural Layer Matrices & Activations Practice Exercises

A binary classification model uses a logistic sigmoid function $\sigma(z) = \frac{1}{1+e^{-z}}$. The decision boundary is defined by $\sigma(z) = 0.5$, where $z = w_1x_1 + w_0$. For a dataset with two features x_1 and x_2, the model's linear component is $z = 2x_1 - x_2 + b$. If the classification boundary passes through the point $(x_1, x_2) = (3, 1)$, what is the value of the bias b?



a. $b = -5$

b. $b = -7$

c. $b = 5$

d. $b = 7$

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Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 **[Intermediate]** • **Target: Neural Layer Matrices & Activations Practice Exercises**

Consider a dataset of observations $\{x_i\}_{i=1}^{n}$ where each x_i is a 2-dimensional vector. The sample covariance matrix for this dataset is given by: $S = (4, -2; -2, 1)$ Principal Component Analysis (PCA) is applied to this dataset. The first principal component corresponds to the direction with the largest variance. Let the direction vector of the first principal component be $\mathbf{v}_1 = (a; b)$. The variance explained by this principal component is given by the corresponding eigenvalue λ_1. If $\mathbf{v}_1$ is a unit vector, what is the value of $a^2 + b^2$?

a. 1

b. 2

c. 4

d. 5

30 **[Intermediate]** • **Target: Neural Layer Matrices & Activations Practice Exercises**

Consider a dataset with two features, X_1 and X_2. The covariance matrix for this dataset is given by: $\Sigma = (10, 5; 5, 20)$ What is the variance of the first feature, X_1?

a. 10

b. 5

c. 20

d. 25

31 **[Intermediate]** • **Target: Neural Layer Matrices & Activations Practice Exercises**

Consider a simple logistic regression model where the decision boundary is defined by the equation $-1.5x_1 + 2x_2 + 5 = 0$. For a data point (x_1, x_2), if $-1.5x_1 + 2x_2 + 5 < 0$, it is classified into Class 0. Otherwise, it is classified into Class 1. Which of the following points lies on the decision boundary?

a. $(-1, 1)$

b. $(2, -2)$

c. $(0, -2.5)$

d. $(3, 0.5)$

32 **[Intermediate]** • **Target: Neural Layer Matrices & Activations Practice Exercises**

Consider a simple linear regression model where the predicted value $\hat{y}$ is given by $\hat{y} = b_0 + b_1 x$. If we have two data points $(x_1, y_1) = (2, 5)$ and $(x_2, y_2) = (4, 9)$, and the Ordinary Least Squares (OLS) method yields the regression line $\hat{y} = 1 + 2x$, what is the predicted value $\hat{y}$ for $x = 3$?

a. 5

b. 6

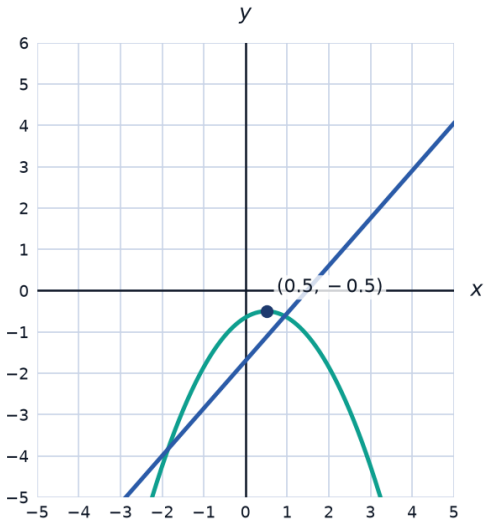
c. 7

d. 8

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33 [Intermediate] • Target: Neural Layer Matrices & Activations Practice Exercises

Consider the quadratic loss function $L(w) = w^2 - 4w + 3$. Find the value of w where the function reaches its minimum.



a. $w = 1$

b. $w = 2$

c. $w = 3$

d. $w = 4$

34 [Intermediate] • Target: Neural Layer Matrices & Activations Practice Exercises

Consider a simple logistic regression model where the decision boundary is defined by the linear equation $w_1x_1 + w_2x_2 + b = 0$. If the weights are $w_1 = 2$ and $w_2 = 3$, and the bias is $b = -7$, what is the equation of the decision boundary?

a. $2x_1 + 3x_2 - 7 = 0$

b. $3x_1 + 2x_2 + 7 = 0$

c. $2x_1 + 3x_2 + 7 = 0$

d. $-2x_1 - 3x_2 + 7 = 0$

35 [Intermediate] • Target: Neural Layer Matrices & Activations Practice Exercises

Consider the following quadratic loss function: $L(w) = 3w^2 - 12w + 10$. What is the value of w that minimizes this loss function?

a. 2

b. 1

c. 4

d. 12

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36 [Intermediate] • Target: Neural Layer Matrices & Activations Practice Exercises

Consider a simple linear regression model where a dependent variable y is predicted from an independent variable x. The model is given by the equation $y = \beta_0 + \beta_1 x + \varepsilon$, where ε is the error term. Suppose we have estimated the regression line to be $\hat{y} = 3 + 2x$. If we observe an independent variable value of $x = 4$, what is the predicted value $\hat{y}$ for the dependent variable?

a. 11

b. 14

c. 10

d. 7

37 [Intermediate] • Target: Neural Layer Matrices & Activations Practice Exercises

Consider a simple quadratic loss function $L(w) = (w^2 - 4w + 5)$. We want to find the minimum of this loss function using gradient descent. The gradient of the loss function with respect to w is given by $\frac{dL}{dw} = 2w - 4$. If we start at an initial weight $w_0 = 5$ and use a learning rate $\alpha = 0.5$, what will be the value of the weight w_1 after the first update step, i.e., $w_1 = w_0 - \alpha \frac{dL}{dw}(w_0)$?

a. 4

b. 3

c. 4.5

d. 5

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Neural Layer Matrices & Activations Practice Exercises • Cross-Lesson: 6.01 Linear Regression and OLS

Consider a simple linear regression model where the predicted value $\hat{y}$ is related to the independent variable x by the equation $\hat{y} = mx + c$. If the line of best fit passes through the points $(1, 5)$ and $(3, 11)$, what is the slope m of this line?

a. 3

b. 2

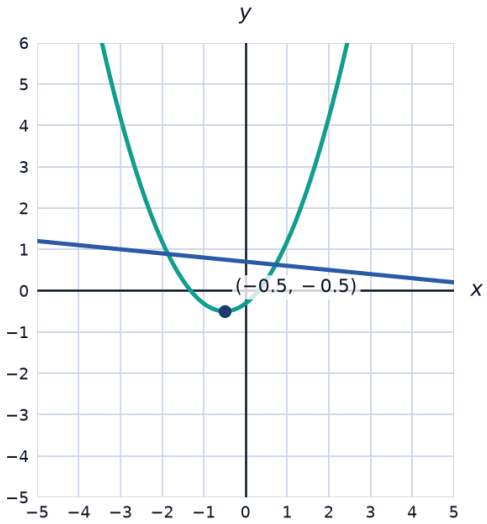
c. 5

d. 6

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39 [Challenging] • Target: Neural Layer Matrices & Activations Practice Exercises • Cross-Lesson: 7.01 Logistic Regression and Classification

Consider a discrete random variable X with the following probability mass function (PMF): $P(X = x) = \frac{x}{10}$ for $x \in \{1, 2, 3, 4\}$. What is the value of $P(X \leq 2)$?



a. $\frac{3}{10}$

b. $\frac{1}{10}$

c. $\frac{6}{10}$

d. $\frac{4}{10}$

40 [Challenging] • Target: Neural Layer Matrices & Activations Practice Exercises • Cross-Lesson: 8.01 PCA and Dimensionality Reduction

Consider a dataset where the number of defective items produced by a machine per hour, Y, follows a Poisson distribution with a rate parameter $\lambda = 3$. The probability mass function (PMF) for a Poisson distribution is given by $P(Y = k) = \frac{\lambda^k e^{-\lambda}}{k!}$ for $k = 0, 1, 2, \ldots$. What is the value of λ in this scenario?

a. $\lambda = 0$

b. $\lambda = 1$

c. $\lambda = 3$

d. $\lambda = e^{-3}$

41 [Challenging] • Target: Neural Layer Matrices & Activations Practice Exercises • Cross-Lesson: 1.01 Vectors and Embeddings in AI

Consider the following quadratic loss function for a regression problem: $L(y, \hat{y}) = (y - \hat{y})^2$. If the predicted value $\hat{y} = 5$ and the true value $y = 3$, what is the value of the loss function?

a. 4

b. -2

c. 9

d. 2

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42 [Challenging] • Target: Neural Layer Matrices & Activations Practice Exercises • Cross-Lesson: 3.01 Gradient Descent and Optimization

Consider two events, A and B, with the following probabilities: $P(A) = 0.4$, $P(B) = 0.5$, and $P(A \cap B) = 0.2$. What is the conditional probability of A given B, denoted as $P(A|B)$?

a. 0.2

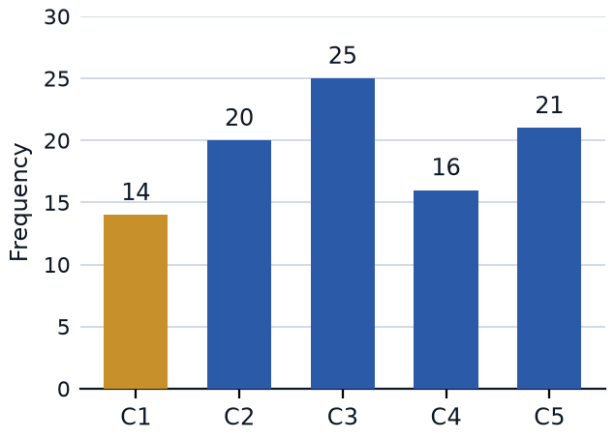
b. 0.4

c. 0.5

d. 0.8

43 [Challenging] • Target: Neural Layer Matrices & Activations Practice Exercises • Cross-Lesson: 4.01 Conditional Probability and Bayes Rule

A centered feature matrix $\mathbf{X}$ has shape $N \times 14$. What is the size and symmetry property of the sample covariance matrix $\mathbf{\Sigma} = \frac{1}{N-1}\mathbf{X}^T\mathbf{X}$?



a. 14×14 symmetric matrix

b. $N \times N$ symmetric matrix

c. 14×14 skew-symmetric matrix

d. $N \times 14$ rectangular matrix

44 [Challenging] • Target: Neural Layer Matrices & Activations Practice Exercises • Cross-Lesson: 5.01 Distributions and Maximum Likelihood

Consider a 2D data point $\mathbf{v} = [3;\ -4]$. What is the squared Euclidean norm (or squared magnitude) of this vector?

a. 7

b. 25

c. 9

d. 16

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45 [Challenging] • Target: Neural Layer Matrices & Activations Practice Exercises • Cross-Lesson: 6.01 Linear Regression and OLS

Given $n = 4$ independent observations from a normal distribution with sum $\sum_{i=1}^{4} x_i = 12$, what is the Maximum Likelihood Estimate (MLE) $\hat{\mu}_{\text{MLE}}$?

a. $\hat{\mu}_{\text{MLE}} = 3$

b. $\hat{\mu}_{\text{MLE}} = 6$

c. $\hat{\mu}_{\text{MLE}} = 12$

d. $\hat{\mu}_{\text{MLE}} = 1.5$

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